

# Appendix D - Per-Engine MMIO Maps

Every memory-mapped device, grouped by subsystem. Each register is 32-bit on the bus unless noted; only the documented low bits carry information.

## D.1 Video chip (\$F0000-\$F049B)

| Address         | Name             | R/W | Notes                                              |
|-----------------|------------------|-----|----------------------------------------------------|
| \$F0000         | VIDEO_CTRL       | R/W | Enable, mode select, sync source.                  |
| \$F0004         | VIDEO_MODE       | R/W | Mode value selector.                               |
| \$F0008         | VIDEO_STATUS     | R   | Bit 0 HAS_CONTENT, bit 1 VBLANK, bit 2 FB_ERR.     |
| \$F000C         | COPPER_CTRL      | R/W | Copper enable.                                     |
| \$F0010         | COPPER_PTR       | R/W | Copper-list base address.                          |
| \$F0014         | COPPER_PC        | R   | Current copper program counter.                    |
| \$F0018         | COPPER_STATUS    | R   | Copper running / stopped.                          |
| \$F001C         | BLT_CTRL         | R/W | Blitter control.                                   |
| \$F0020         | BLT_OP           | R/W | Blitter operation.                                 |
| \$F0024         | BLT_SRC          | R/W | Source address.                                    |
| \$F0028         | BLT_DST          | R/W | Destination address.                               |
| \$F002C         | BLT_WIDTH        | R/W | Width, byte count for MEMCOPY, or packed endpoint. |
| \$F0030         | BLT_HEIGHT       | R/W | Height or packed endpoint.                         |
| \$F0034         | BLT_SRC_STRIDE   | R/W | Source stride.                                     |
| \$F0038         | BLT_DST_STRIDE   | R/W | Destination stride.                                |
| \$F003C         | BLT_COLOR        | R/W | Fill colour or scale destination size.             |
| \$F0040         | BLT_MASK         | R/W | Mask address.                                      |
| \$F0044         | BLT_STATUS       | R   | Blitter busy / done.                               |
| \$F0048-\$F0054 | VIDEO_RASTER_*   | R/W | Raster split lines and per-line palette swaps.     |
| \$F0058-\$F0074 | BLT_MODE7_*      | R/W | Mode 7 affine texture registers.                   |
| \$F0078         | VIDEO_PAL_INDEX  | R/W | CLUT palette write index.                          |
| \$F007C         | VIDEO_PAL_DATA   | R/W | CLUT palette data.                                 |
| \$F0080         | VIDEO_COLOR_MODE | R/W | 0 RGBA32, 1 CLUT8.                                 |
| \$F0084         | VIDEO_FB_BASE    | R/W | Framebuffer base address.                          |
| \$F0088-\$F0487 | palette          | R/W | 256-entry direct palette table.                    |
| \$F0488-\$F049B | BLT_EXT_*        | R/W | Extended blitter (large modes).                    |

VIDEO\_MODE values:

| Value | Frame size   |
|-------|--------------|
| \$00  | 640 by 480   |
| \$01  | 800 by 600   |
| \$02  | 1024 by 768  |
| \$03  | 1280 by 960  |
| \$04  | 320 by 200   |
| \$05  | 320 by 240   |
| \$06  | 1920 by 1080 |
| \$07  | 960 by 540   |

BLT\_OP values:

| Value | Operation    |
|-------|--------------|
| 0     | COPY         |
| 1     | FILL         |
| 2     | LINE         |
| 3     | MASKED_COPY  |
| 4     | ALPHA_COPY   |
| 5     | MODE7        |
| 6     | COLOR_EXPAND |
| 7     | SCALE        |
| 8     | MEMCOPY      |

For MEMCOPY, BLT\_SRC is the source byte address, BLT\_DST is the destination byte address, and BLT\_WIDTH is the byte count.

## D.2 Terminal / serial / input (\$F0700-\$F07FF)

| Address | Name             | R/W | Notes                                  |
|---------|------------------|-----|----------------------------------------|
| \$F0700 | TERM_OUT         | W   | Print one byte.                        |
| \$F0704 | TERM_STATUS      | R   | Bit 0 input avail, bit 1 output ready. |
| \$F0708 | TERM_IN          | R   | Dequeue cooked input byte.             |
| \$F070C | TERM_LINE_STATUS | R   | Bit 0 complete line in queue.          |
| \$F0710 | TERM_ECHO        | R/W | Local echo.                            |
| \$F0724 | TERM_CTRL        | R/W | Bit 0 line-input mode.                 |
| \$F0728 | TERM_KEY_IN      | R   | Dequeue raw key.                       |
| \$F072C | TERM_KEY_STATUS  | R   | Bit 0 raw key avail.                   |

| Address | Name             | R/W | Notes                                                      |
|---------|------------------|-----|------------------------------------------------------------|
| \$F0730 | MOUSE_X          | R   | Absolute X (low 16 bits).                                  |
| \$F0734 | MOUSE_Y          | R   | Absolute Y.                                                |
| \$F0738 | MOUSE_BUTTONS    | R   | Bit 0 L, 1 R, 2 M.                                         |
| \$F073C | MOUSE_STATUS     | R   | Bit 0 changed (clears on read).                            |
| \$F0740 | SCAN_CODE        | R   | Dequeue raw scancode.                                      |
| \$F0744 | SCAN_STATUS      | R   | Bit 0 scancode avail.                                      |
| \$F0748 | SCAN_MODIFIERS   | R   | Bit 0 shift, 1 ctrl, 2 alt, 3 caps.                        |
| \$F074C | MOUSE_CTRL       | R/W | Bit 0 request relative mode.                               |
| \$F0750 | RTC_EPOCH        | R   | Seconds since 1970-01-01 00:00:00 UTC.                     |
| \$F0754 | MOUSE_DX         | R   | Signed accumulated dX (clears on read).                    |
| \$F0758 | MOUSE_DY         | R   | Signed accumulated dY.                                     |
| \$F075C | RTC_MONO_USEC_LO | R   | Low 32 bits of monotonic microseconds since engine start.  |
| \$F0760 | RTC_MONO_USEC_HI | R   | High 32 bits of monotonic microseconds since engine start. |
| \$F07F0 | TERM_SENTINEL    | W   | Write \$DEAD to stop CPU.                                  |

## D.3 SoundChip (\$F0800-\$F0B7F)

| Range           | Block           | Notes                                             |
|-----------------|-----------------|---------------------------------------------------|
| \$F0800-\$F08FF | Global          | AUDIO_CTRL, ENV_SHAPE, filter, reverb, overdrive. |
| \$F0900-\$F093F | Square          | SQ_FREQ, SQ_DUTY, SQ_ENV_*, SQ_GATE.              |
| \$F0940-\$F097F | Triangle        | Same shape.                                       |
| \$F0980-\$F09BF | Sine            | Same shape.                                       |
| \$F09C0-\$F09FF | Noise           | LFSR period, volume, gate.                        |
| \$F0A00-\$F0A1F | Sync / ring-mod | Source-channel selects.                           |
| \$F0A20-\$F0A6F | Sawtooth        | Same shape.                                       |
| \$F0A80-\$F0B7F | Flex 0-3        | 64-byte per-channel block.                        |

Audio player control blocks in the same area:

| Range           | Block    | Registers                                                                                                                                                          |
|-----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$F0B80-\$F0B91 | AHX      | AHX_PLUS_CTRL, AHX_PLAY_PTR, AHX_PLAY_LEN, AHX_PLAY_CTRL, AHX_PLAY_STATUS, AHX_SUBSONG.                                                                            |
| \$F0BA0-\$F0BBF | MIDI/MUS | MIDI_PLAY_PTR, MIDI_PLAY_LEN, MIDI_PLAY_CTRL, MIDI_PLAY_STATUS (bit 0 busy, bit 1 error, bit 2 paused, bit 3 loading), MIDI_POSITION, MIDI_VOLUME, MIDI_TEMPO_BPM. |
| \$F0BC0-\$F0BD7 | MOD      | MOD_PLAY_PTR, MOD_PLAY_LEN, MOD_PLAY_CTRL, MOD_PLAY_STATUS, MOD_FILTER_MODEL, MOD_POSITION.                                                                        |

| Range           | Block | Registers                                                                                                                                           |
|-----------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| \$F0BD8-\$F0BF3 | WAV   | WAV_PLAY_PTR, WAV_PLAY_LEN, WAV_PLAY_CTRL, WAV_PLAY_STATUS, WAV_POSITION, WAV_PLAY_PTR_HI, WAV_CHANNEL_BASE, WAV_VOLUME_L, WAV_VOLUME_R, WAV_FLAGS. |

## D.4 SFX triggers (\$F0E80-\$F0EFF)

| Per-channel offset | Field        | Notes                      |
|--------------------|--------------|----------------------------|
| +\$00              | SFX_PTR      | Sample base address.       |
| +\$04              | SFX_LEN      | Sample length in bytes.    |
| +\$08              | SFX_LOOP_PTR | Loop point.                |
| +\$0C              | SFX_LOOP_LEN | Loop length.               |
| +\$10              | SFX_FREQ     | Playback rate.             |
| +\$14              | SFX_VOL      | Volume (0-65535).          |
| +\$18              | SFX_FORMAT   | Sample format.             |
| +\$1C              | SFX_CTRL     | Trigger / one-shot / loop. |

Four channels at \$F0E80, \$F0EA0, \$F0EC0, \$F0EE0.

## D.5 PSG / AY-3-8910 (\$F0C00-\$F0C20)

| Offset      | Register                                             |
|-------------|------------------------------------------------------|
| +\$00-+\$0F | R0-R15, one byte per register (byte-contiguous).     |
| +\$10       | PSG_PLAY_PTR (32-bit).                               |
| +\$14       | PSG_PLAY_LEN (32-bit).                               |
| +\$18       | PSG_PLAY_CTRL (bit 0 start, bit 1 stop, bit 2 loop). |
| +\$1C       | PSG_PLAY_STATUS (bit 0 busy, bit 1 error).           |
| +\$20       | PSG_PLUS_CTRL.                                       |

## D.6 SN76489 (\$F0C30-\$F0C3F)

| Offset | Register       |
|--------|----------------|
| +\$00  | SN_PORT_WRITE. |
| +\$01  | SN_PORT_READY. |
| +\$02  | SN_PORT_MODE.  |

## D.7 SID family

Each SID instance is a byte-contiguous block in the original 6581/8580 register layout. Three independent instances live at three bases:

| Block | Range           | Notes                   |
|-------|-----------------|-------------------------|
| SID   | \$F0E00-\$F0E1C | Primary SID.            |
| SID2  | \$F0E30-\$F0E4C | Second independent SID. |
| SID3  | \$F0E50-\$F0E6C | Third independent SID.  |

Within each instance the offsets follow the original chip layout, one byte per register:

| Offset      | Register                                                                                                                                                 |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| +\$00-+\$06 | Voice 1: <code>FREQ_L0</code> , <code>FREQ_HI</code> , <code>PW_L0</code> , <code>PW_HI</code> , <code>CTRL</code> , <code>AD</code> , <code>SR</code> . |
| +\$07-+\$0D | Voice 2 (same shape).                                                                                                                                    |
| +\$0E-+\$14 | Voice 3 (same shape).                                                                                                                                    |
| +\$15-+\$17 | Filter: <code>FC_L0</code> , <code>FC_HI</code> , <code>RES_FILT</code> .                                                                                |
| +\$18       | <code>MODE_VOL</code> .                                                                                                                                  |
| +\$19       | SID+ control ( <code>SID_PLUS_CTRL</code> ) / <code>POT_X</code> mirror.                                                                                 |
| +\$1A       | <code>POT_Y</code> .                                                                                                                                     |
| +\$1B       | <code>OSC3</code> .                                                                                                                                      |
| +\$1C       | <code>ENV3</code> .                                                                                                                                      |

The primary SID block at \$F0E00 has an attached SID file player block after the voice registers:

| Offset (from \$F0E00) | Register                                          |
|-----------------------|---------------------------------------------------|
| +\$20                 | <code>SID_PLAY_PTR</code> (32-bit).               |
| +\$24                 | <code>SID_PLAY_LEN</code> (32-bit).               |
| +\$28                 | <code>SID_PLAY_CTRL</code> (start / stop / loop). |
| +\$2C                 | <code>SID_PLAY_STATUS</code> (busy / error).      |
| +\$2D                 | <code>SID_SUBSONG</code> .                        |

The two flexible audio blocks \$F0C40-\$F0CFF and \$F0D40-\$F0DFF are not SID2 and SID3. They are extra-channel mirrors (channels 4-6 and 7-9 in the global SoundChip channel space) and have the flex-channel layout documented under SoundChip (D.3), not the SID layout. A program that wants SID2 / SID3 should program them at \$F0E30 and \$F0E50.

## D.8 POKEY (\$F0D00-\$F0D20)

| Offset | Register                   |
|--------|----------------------------|
| +\$00  | <code>POKEY_AUDF1</code> . |
| +\$01  | <code>POKEY_AUDC1</code> . |
| +\$02  | <code>POKEY_AUDF2</code> . |
| +\$03  | <code>POKEY_AUDC2</code> . |
| +\$04  | <code>POKEY_AUDF3</code> . |
| +\$05  | <code>POKEY_AUDC3</code> . |

| Offset | Register             |
|--------|----------------------|
| +\$06  | POKEY_AUDF4.         |
| +\$07  | POKEY_AUDC4.         |
| +\$08  | POKEY_AUDCTL.        |
| +\$09  | POKEY_PLUS_CTRL.     |
| +\$0A  | POKEY_RANDOM (read). |
| +\$10  | SAP_PLAY_PTR.        |
| +\$14  | SAP_PLAY_LEN.        |
| +\$18  | SAP_PLAY_CTRL.       |
| +\$1C  | SAP_PLAY_STATUS.     |
| +\$20  | SAP_SUBSONG.         |

## D.9 TED audio (\$F0F00-\$F0F1F)

| Offset | Register         |
|--------|------------------|
| +\$00  | TED_FREQ1_L0.    |
| +\$01  | TED_FREQ2_L0.    |
| +\$02  | TED_FREQ2_HI.    |
| +\$03  | TED_SND_CTRL.    |
| +\$04  | TED_FREQ1_HI.    |
| +\$05  | TED_PLUS_CTRL.   |
| +\$10  | TED_PLAY_PTR.    |
| +\$14  | TED_PLAY_LEN.    |
| +\$18  | TED_PLAY_CTRL.   |
| +\$1C  | TED_PLAY_STATUS. |

## D.10 TED video (\$F0F20-\$F0F6B)

40 by 25 text with 121-colour cells, raster position registers, raster compare registers, and raster pending status; see Chapter 6.

## D.11 VGA (\$F1000-\$F13FF)

| Range           | Subsystem                       |
|-----------------|---------------------------------|
| \$F1000-\$F1008 | VGA_MODE, VGA_STATUS, VGA_CTRL. |
| \$F1010-\$F1018 | Sequencer (VGA_SEQ_*).          |
| \$F1020-\$F102C | CRTC (VGA_CRTC_*).              |

| Range           | Subsystem                          |
|-----------------|------------------------------------|
| \$F1030-\$F103C | Graphics controller (VGA_GC_*).    |
| \$F1040-\$F1044 | Attribute controller (VGA_ATTR_*). |
| \$F1050-\$F105C | DAC + palette index/data.          |
| \$F1100-\$F13FF | Palette RAM.                       |

## D.12 ULA (\$F2000-\$F2017)

| Offset | Register               |
|--------|------------------------|
| +\$00  | ULA_BORDER (bits 0-2). |
| +\$04  | ULA_CTRL.              |
| +\$08  | ULA_STATUS.            |
| +\$0C  | ULA_ADDR_LO.           |
| +\$10  | ULA_ADDR_HI.           |
| +\$14  | ULA_DATA.              |

ULA VRAM aperture at \$FA000-\$FBAFF.

## D.13 ANTIC + GTIA (\$F2100-\$F21FB)

ANTIC block at \$F2100-\$F213F with DMACTL, CHACTL, DLISTL/H, HSCR0L, VSCR0L, PMBASE, CHBASE, WSYNC, VCOUNT, NMIEI/IST/RES. GTIA at \$F2140-\$F21FB with COLPM0-3, COLPF0-3, COLBK, PRIOR, GRACLT, CONSOL, player/missile positions, sizes, graphics bytes, collision latches, and HITCLR.

## D.14 File I/O (\$F2200-\$F221F)

Used by LOAD, SAVE, BLOAD, COMPILE, TRANSPILE, ASSEMBLE, DIR, and TYPE, and by machine code that talks to the disk volume directly.

| Offset | Register                                                                                                                    |
|--------|-----------------------------------------------------------------------------------------------------------------------------|
| +\$00  | FILE_NAME_PTR.                                                                                                              |
| +\$04  | FILE_DATA_PTR.                                                                                                              |
| +\$08  | FILE_DATA_LEN (write byte count; ignored by read).                                                                          |
| +\$0C  | FILE_CTRL (write triggers).                                                                                                 |
| +\$10  | FILE_STATUS.                                                                                                                |
| +\$14  | FILE_RESULT_LEN (actual read/list byte count; 0 after accepted-path read failure).                                          |
| +\$18  | FILE_ERROR_CODE.                                                                                                            |
| +\$1C  | FILE_READ_MAX (one-shot read cap in bytes; larger file refused with FILE_ERR_RANGE before any byte is copied; 0 unbounded). |

FILE\_ERROR\_CODE values:

| Value | Meaning                                                                                         |
|-------|-------------------------------------------------------------------------------------------------|
| 0     | OK.                                                                                             |
| 1     | Not found.                                                                                      |
| 2     | Permission.                                                                                     |
| 3     | Path traversal.                                                                                 |
| 4     | Range error: staged data would reach \$FFFF0000, wrap the 32-bit pointer, or exceed active RAM. |

## D.15 Amiga Paula DMA (\$F2260-\$F22AF)

Four-channel DMA sample engine. Each channel is 16 bytes:

| Offset | Register             |
|--------|----------------------|
| +\$00  | PTR sample pointer.  |
| +\$04  | LEN length in words. |
| +\$08  | PER period.          |
| +\$0C  | VOL volume.          |

Global registers:

| Address | Register         |
|---------|------------------|
| \$F22A0 | AROS_AUD_DMACON. |
| \$F22A4 | AROS_AUD_STATUS. |
| \$F22A8 | AROS_AUD_INTENA. |

## D.16 Media loader (\$F2300-\$F231F)

Dispatches by file extension into the matching audio engine. Used by SOUND PLAY (Chapter 23).

| Offset | Register                                                                                   |
|--------|--------------------------------------------------------------------------------------------|
| +\$00  | MEDIA_NAME_PTR.                                                                            |
| +\$04  | MEDIA_SUBSONG.                                                                             |
| +\$08  | MEDIA_CTRL (1 play, 2 stop).                                                               |
| +\$0C  | MEDIA_STATUS (0 idle, 1 loading, 2 playing, 3 error).                                      |
| +\$10  | MEDIA_TYPE (1 SID, 2 PSG, 3 TED, 4 AHX, 5 POKEY, 6 MOD, 7 WAV, 8 MIDI/MUS).                |
| +\$14  | MEDIA_ERROR (0 OK, 1 not found, 2 bad format, 3 unsupported, 4 name invalid, 5 too large). |

## D.17 RUN loader block (\$F2320-\$F233F)

The RUN keyword uses this block for service launches. It is not a normal reader programming route; use RUN syntax, COCALL, or IE Mon as described in the main chapters.



| Offset | Field         |
|--------|---------------|
| +\$00  | name pointer. |
| +\$04  | control.      |
| +\$08  | status.       |
| +\$0C  | type.         |
| +\$10  | error.        |
| +\$14  | session.      |

## D.18 Coprocessor (\$F2340-\$F238F)

COSTART, COSTOP, COCALL, COSTATUS, and COWAIT use this command block:

| Offset | Register                  |
|--------|---------------------------|
| +\$00  | COPROC_CMD.               |
| +\$04  | COPROC_CPU_TYPE.          |
| +\$08  | COPROC_CMD_STATUS.        |
| +\$0C  | COPROC_CMD_ERROR.         |
| +\$10  | COPROC_TICKET.            |
| +\$14  | COPROC_TICKET_STATUS.     |
| +\$18  | COPROC_OP.                |
| +\$1C  | COPROC_REQ_PTR.           |
| +\$20  | COPROC_REQ_LEN.           |
| +\$24  | COPROC_RESP_PTR.          |
| +\$28  | COPROC_RESP_CAP.          |
| +\$2C  | COPROC_TIMEOUT.           |
| +\$30  | COPROC_NAME_PTR.          |
| +\$34  | COPROC_WORKER_STATE.      |
| +\$38  | COPROC_STATS_OPS.         |
| +\$3C  | COPROC_STATS_BYTES.       |
| +\$40  | COPROC_IRQ_CTRL.          |
| +\$44  | COPROC_DISPATCH_OVERHEAD. |
| +\$48  | COPROC_COMPLETED_TICKET.  |

Extended monitor block (\$F23B0-\$F23BF):

| Offset | Register              |
|--------|-----------------------|
| +\$00  | COPROC_RING_DEPTH.    |
| +\$04  | COPROC_WORKER_UPTIME. |

| Offset | Register            |
|--------|---------------------|
| +\$08  | COPROC_STATS_RESET. |
| +\$0C  | COPROC_BUSY_PCT.    |

## D.19 IRQ diagnostics (\$F23C0-\$F23DF)

| Offset | Register                |
|--------|-------------------------|
| +\$00  | In-service mask.        |
| +\$04  | State flags.            |
| +\$08  | Pending mask.           |
| +\$0C  | Delivered counters.     |
| +\$10  | Blocked counters.       |
| +\$14  | RTE count.              |
| +\$18  | Consecutive STOP spins. |
| +\$1C  | Watchdog event count.   |

## D.20 SysInfo (\$F2400-\$F24FF)

| Offset | Register               |
|--------|------------------------|
| +\$00  | SYSINFO_TOTAL_RAM_LO.  |
| +\$04  | SYSINFO_TOTAL_RAM_HI.  |
| +\$08  | SYSINFO_ACTIVE_RAM_LO. |
| +\$0C  | SYSINFO_ACTIVE_RAM_HI. |

## D.21 HOST appliance block (\$F1400-\$F140F)

| Offset | Name    | Width  | Notes                                                                                                                     |
|--------|---------|--------|---------------------------------------------------------------------------------------------------------------------------|
| +\$00  | command | byte   | Subverb enum (W).                                                                                                         |
| +\$04  | trigger | byte   | Non-zero fires the command (W).                                                                                           |
| +\$08  | status  | byte   | Terminal-state enum (R).                                                                                                  |
| +\$0C  | exit    | 32-bit | Full 32-bit exit code from the underlying action (R); byte lanes at +\$0C-\$0F return successive bytes of the same value. |

See Chapter 36 for the subverb enum and the state machine. Reachable from IE64, IE32, M68K, and x86; not reachable from the 6502 or Z80.

## D.22 Voodoo 3D (\$F8000-\$F87FF)

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Status, framebuffer base, clip rect, triangle setup, texture descriptors, fog, alpha, chroma-key, Z-buffer. Documented in Chapter 9. Texture RAM at \$D0000-\$DFFFF.